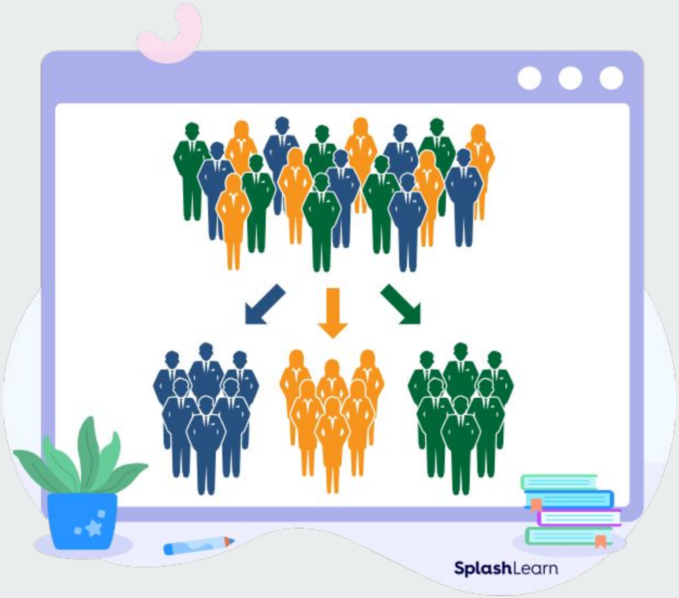
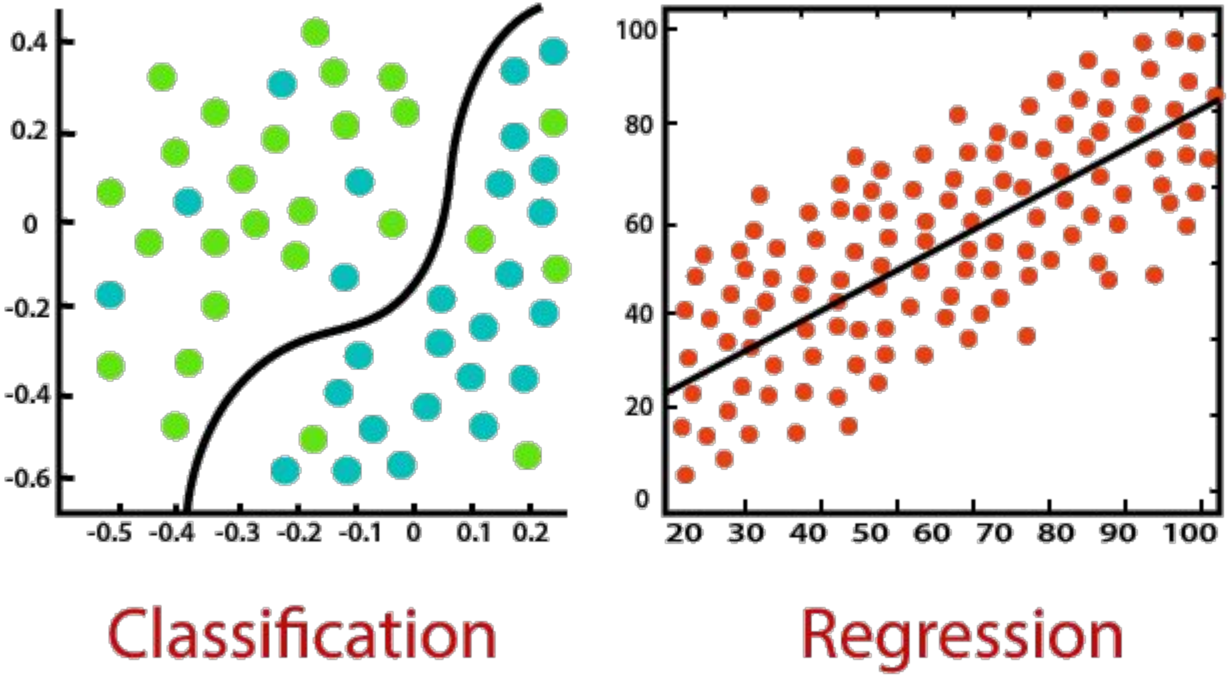
Classification



Classification

- **Binary** ← Els propers dies treballarem amb aquest.
- **Multiclass**

Classification vs Regression



Binary Classification

- El model prediu una de dues classes possibles.
  - E.g., positiu vs. negatiu, 1 vs. 0, yes vs. no.
- Exemples:
  - Email: spam o no spam.
  - Diagnosis: càncer or no càncer.
  - Review: positiu o negatiu.
  - Reserva hotel: cancel·lada o no.

Binary Classification

- Realment, els models normalment prediuen una probabilitat.
- Per exemple, donat una fila (mostra), la probabilitat que aquesta pertanyi a la classe 1.
    - Llavors prediu classe 1 si la probabilitat és > 50% o sinó classe 0.

Binary Classification

- Confusion matrix (matriu de confusió):
    - **True Positives (TP)**: Positius predits correctament.

Binary Classification

- Confusion matrix (matriu de confusió):
  - **True Positives (TP)**: Positius predits correctament.
  - **True Negatives (TN)**: Negatius predits correctament.

Binary Classification

- Confusion matrix (matriu de confusió):
    - **True Positives (TP)**: Positius predits correctament.
    - **True Negatives (TN)**: Negatius predits correctament.
    - **False Positives (FP)**: Positius predits incorrectament. Realment han de ser negatius.

Binary Classification

- Confusion matrix (matriu de confusió):
    - **True Positives (TP)**: Positius predits correctament.
    - **True Negatives (TN)**: Negatius predits correctament.
    - **False Positives (FP)**: Positius predits incorrectament. Realment han de ser negatius.
    - **False Negatives (FN)**: Negatius predits incorrectament. Realment han de ser positius.

Classification problem

Outlook	Temperature	Humidity	Windy	Play golf (y)
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes

Classification predictions

Play golf (y)	Model (ŷ)
No	No
No	Yes
Yes	Yes
Yes	Yes
Yes	No
No	No
Yes	Yes
No	Yes
Yes	Yes

Classification - Confusion matrix

Play golf (y)	Model (ŷ)
No	No
No	Yes
Yes	Yes
Yes	Yes
Yes	No
No	No
Yes	Yes
No	Yes
Yes	Yes

		Predicted class	
		Positive	Negative
True class	Positive		
	Negative		

Classification - Confusion matrix

Play golf (y)	Model (ŷ)
No	No
No	Yes
Yes	Yes
Yes	Yes
Yes	No
No	No
Yes	Yes
No	Yes
Yes	Yes

		Predicted class	
		Positive	Negative
True class	Positive	4	
	Negative		

Classification - Confusion matrix

Play golf (y)	Model (ŷ)
No	No
No	Yes
Yes	Yes
Yes	Yes
Yes	No
No	No
Yes	Yes
No	Yes
Yes	Yes

		Predicted class	
		Positive	Negative
True class	Positive	4	1
	Negative		

Classification - Confusion matrix

Play golf (y)	Model (ŷ)
No	No
No	Yes
Yes	Yes
Yes	Yes
Yes	No
No	No
Yes	Yes
No	Yes
Yes	Yes

		Predicted class	
		Positive	**Negative**
True class	**Positive**	4	1
	Negative	2	

Classification - Confusion matrix

Play golf (y)	Model (ŷ)
No	No
No	Yes
Yes	Yes
Yes	Yes
Yes	No
No	No
Yes	Yes
No	Yes
Yes	Yes

		Predicted class	
		Positive	Negative
True class	Positive	4	1
	Negative	2	2

Classification - Confusion matrix

Play golf (y)	Model (ŷ)
No	No
No	Yes
Yes	Yes
Yes	Yes
Yes	No
No	No
Yes	Yes
No	Yes
Yes	Yes

		Predicted class	
		Positive	Negative
True class	Positive	TP = 4	FN = 1
	Negative	FP = 2	TN = 2

Classification - Metrics

- **Accuracy**:
  - Medeix la proporció de prediccions correctes (ja siguin positives o negatives).
  - Mètrica molt comú en classificació.
  - Contres:
    - Enganyós quan tenim desbalanceig de classes.
    - No mostra com de bé va el model predint cada una de les classes.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Classification - Metrics

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

<table>
<tr><td colspan="2"></td><td colspan="2">Predicted class</td></tr>
<tr><td colspan="2"></td><td>Positive</td><td>Negative</td></tr>
<tr><td rowspan="2">True class</td><td>Positive</td><td>TP = 4</td><td>FN = 1</td></tr>
<tr><td>Negative</td><td>FP = 2</td><td>TN = 2</td></tr>
</table>

Classification - Metrics

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{4+2}{4+2+2+1} = \frac{6}{9} = 66.7\%$$

		Predicted class	
		Positive	**Negative**
True class	**Positive**	TP = 4	FN = 1
	Negative	FP = 2	TN = 2

Classification - Metrics

- **Recall:**
  - També conegut com True Positive Rate (TPR) o Sensitivity.
  - Medeix la proporció de positius (de tots els del dataset) que el model prediu correctament.
  - Bona mètrica quan “fallar” la predicció de positius té un cost alt.
    - E.g., salut, detecció de frau.
  - Cons: un model dummy que sempre prediu classe 1 té un Recall de 100%.

$$\mathrm{Recall} = \mathrm{TPR} = \frac{TP}{TP + FN}$$

Classification - Metrics

$$\text{Recall} = \text{TPR} = \frac{TP}{TP + FN}$$

		Predicted class	
		Positive	**Negative**
True class	**Positive**	TP = 4	FN = 1
	Negative	FP = 2	TN = 2

Classification - Metrics

$$\text{Recall} = \text{TPR} = \frac{TP}{TP + FN} = \frac{4}{4+1} = \frac{4}{5} = 80\%$$

		Predicted class	
		Positive	**Negative**
True class	**Positive**	TP = 4	FN = 1
	Negative	FP = 2	TN = 2

Classification - Metrics

- **Precision:**
  - Medeix la proporció de prediccions positives que ha fet el model que són correctes.
  - Bona mètrica quan el cost d'un fals positiu és alt.
    - E.g., minimitzar el nombre de no-spam incorrectament classificats com spam.
  - Cons: un model dummy que correctament prediu una sola fila com a classe 1 té Precision de 100%.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Classification - Metrics

$$\text{Precision} = \frac{TP}{TP + FP}$$

		Predicted class	
		Positive	**Negative**
True class	**Positive**	TP = 4	FN = 1
	Negative	FP = 2	TN = 2

Classification - Metrics

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{4}{4+2} = \frac{4}{6} = 66.7\%$$

		Predicted class	
		Positive	**Negative**
True class	**Positive**	TP = 4	FN = 1
	Negative	FP = 2	TN = 2

Classification - Metrics

- **Balanced Accuracy:**
  - Medeix la proporció de prediccions correctes (ja siguin positives o negatives).
  - Útil en datasets amb target desbalancejat (imbalanced).
  - No mostra com de bé va el model predint cada una de les classes.

$$\text{Balanced Accuracy} = \frac{TPR + TNR}{2}$$

Classification - Metrics

$$\text{Balanced Accuracy} = \frac{TPR + TNR}{2}$$

		Predicted class	
		Positive	Negative
True class	Positive	TP = 4	FN = 1
	Negative	FP = 2	TN = 2

Classification - Metrics

$$\text{Balanced Accuracy} = \frac{TPR + TNR}{2} = \frac{\frac{4}{5} + \frac{2}{4}}{2} = \frac{0.8 + 0.5}{2} = 65\%$$

		Predicted class	
		Positive	**Negative**
True class	**Positive**	TP = 4	FN = 1
	Negative	FP = 2	TN = 2

Classification - Metrics: Play Golf

- Accuracy = 0.667 = 66.7%
- Balanced Accuracy = 0.650 = 65%
- Recall = 0.800 = 80%
- Precision = 0.667 = 66.7%

Classification - Misleading Accuracy

100 patients:

99 without cancer

1 with cancer

Great model idea:

Always predict no cancer

Classification - Misleading Accuracy

100 patients:
    99 without cancer
    1 with cancer

Great model idea:
    Always predict no cancer

		Predicted class	
		Positive	**Negative**
True class	**Positive**	TP = 0	FN = 1
	Negative	FP = 0	TN = 99

Classification - Misleading Accuracy

100 patients:
- 99 without cancer
- 1 with cancer

Great model idea:
Always predict no cancer

		Predicted class	
		Positive	**Negative**
True class	**Positive**	TP = 0	FN = 1
	Negative	FP = 0	TN = 99

$$\text{Accuracy} = \frac{99}{100} = 99\%$$

Classification - Misleading Accuracy

100 patients:
- 99 without cancer
- 1 with cancer

Great model idea:
Always predict no cancer
Not great at all...

		Predicted class	
		Positive	**Negative**
True class	**Positive**	TP = 0	FN = 1
	Negative	FP = 0	TN = 99

Accuracy = 99%

Balanced Accuracy = 49.5%

Recall = 0%

Precision = NaN

Classification - More Metrics

https://en.wikipedia.org/wiki/Precision_and_recall

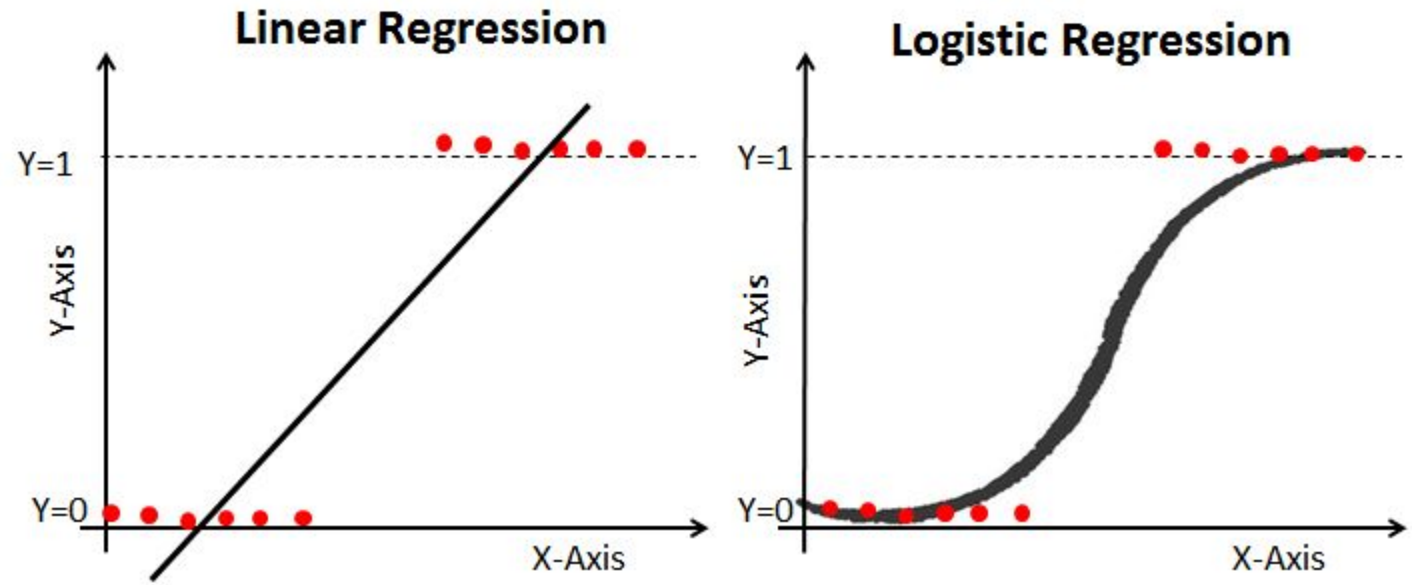
Stratification

Stratification

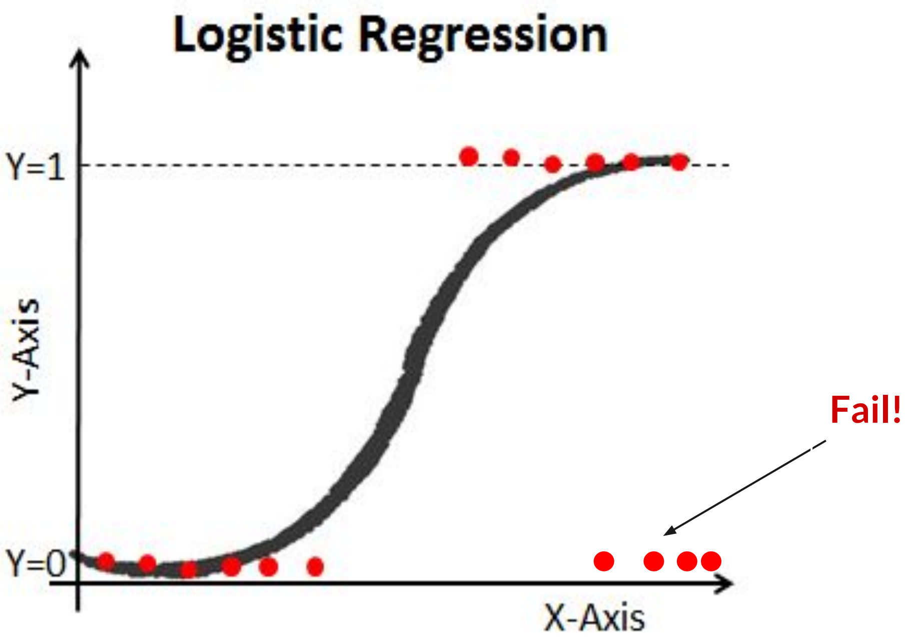
- Tècnica per assegurar que cada classe estigui representada de manera apropiada.
- És especialment important quan el conjunt de dades està desequilibrat (imbalanced).
- S'ha d'aplicar tant en la divisió entre train/test com en cross-validation.
    - Garanteix que train/test/validation tinguin la mateixa proporció de la classe objectiu.

Logistic Regression

Logistic Regression



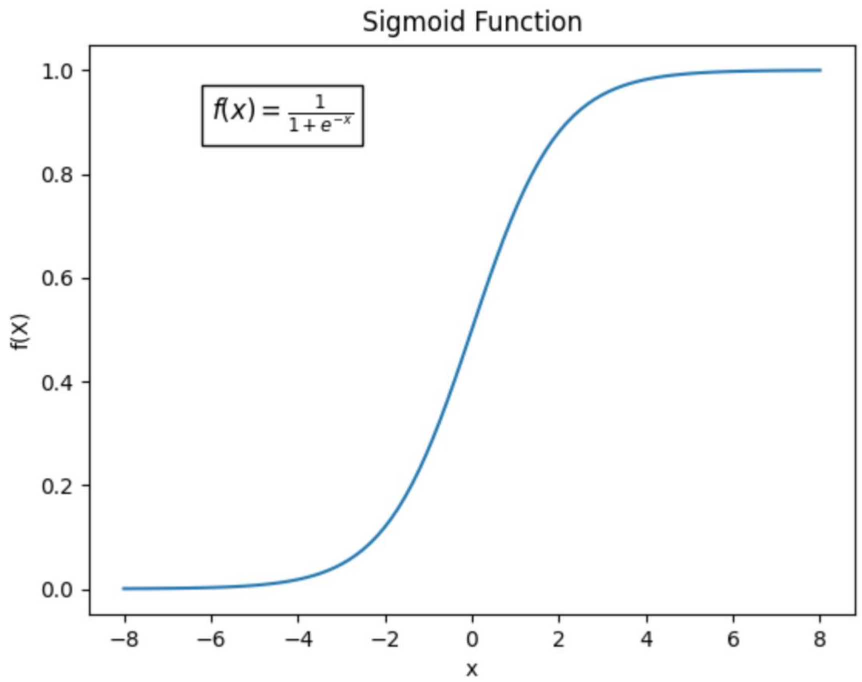
Logistic Regression – Non-Linear



Logistic Regression

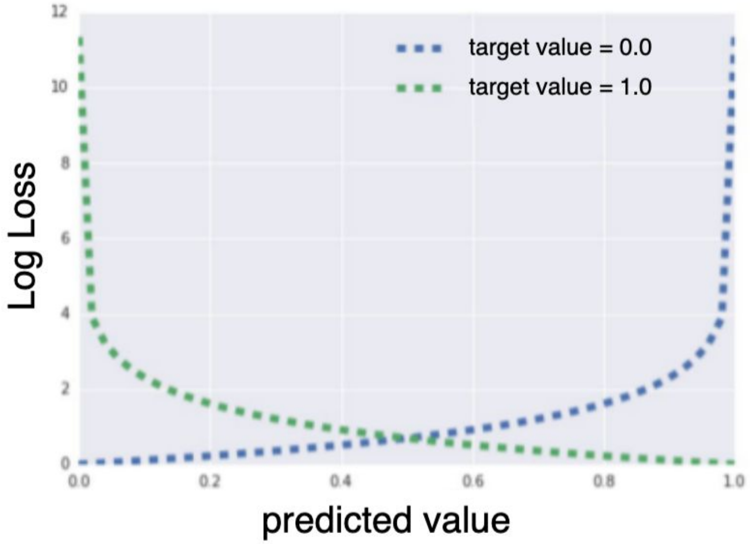
$$P(y=1|X) = \text{sigmoid}(z) = \frac{1}{1+e^{-z}}$$

$$z = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \ldots + \hat{\beta}_k x_k$$



Minimize Log-Loss

$$LogLoss = \sum_{(x,y)\in D} -y\,log(y') - (1-y)\,log(1-y')$$



Logistic Regression: Penalty & C

- L1: $$J(w) = C \cdot \mathrm{LogLoss}(w) + \sum_{j=1}^{n} |w_j|$$
- L2: $$J(w) = C \cdot \mathrm{LogLoss}(w) + \frac{1}{2} \sum_{j=1}^{n} w_j^2$$
- EN: $$J(w) = C \cdot \mathrm{LogLoss}(w) + \mathrm{EN}(\mathrm{w})$$

Logistic Regression: Important Hyperparams.

- LogisticRegressor (sklearn)
- **penalty**: None, l1, l2, elasticnet
- **C**
- **solver**

Logistic Regression: Important Hyperparams.

- C:
  - Hiperparàmetre de regularització.
  - Només s'utilitza quan el paràmetre penalty no és "None".
  - Un valor de C gran pot conduir a overfitting.
  - Un valor de C petit pot conduir a underfitting.
  - Els valors típics oscil·len entre 0.001 i 1000.

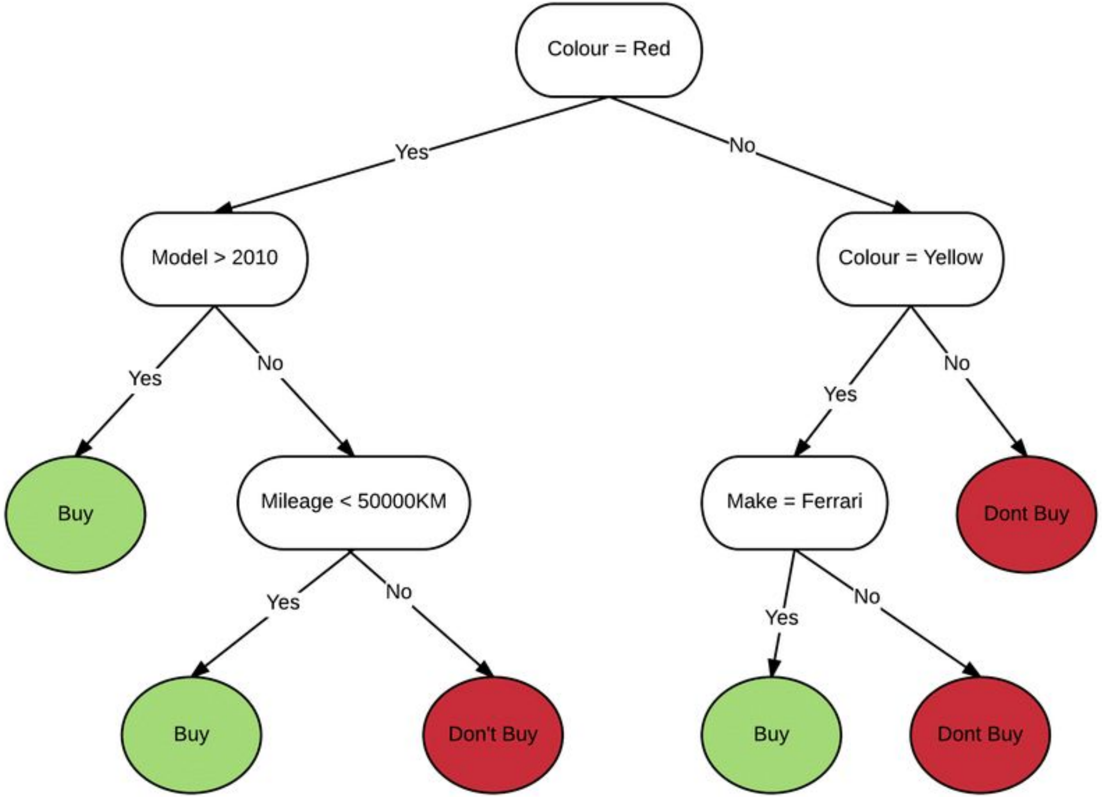
Logistic Regression: Important Hyperparams.

- **solver**:
    - Algorisme utilitzat per a l'optimització del model.
    - Diferents solvers poden gestionar diferents tipus de dades.
    - Diferents solvers tenen característiques de rendiment i eficiència computacional diferents.
    - Consulta la documentació per veure les opcions i les seves característiques.
        - https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html

Decision Tree

DT

- DecisionTreeClassifier (sklearn)



Impurity Criterion

Gini Index

$$I_G = 1 - \sum_{j=1}^{c} p_j^2$$

p_j: proportion of the samples that belongs to class c for a particular node

Entropy

$$I_H = - \sum_{j=1}^{c} p_j log_2(p_j)$$

p_j: proportion of the samples that belongs to class c for a particular node.

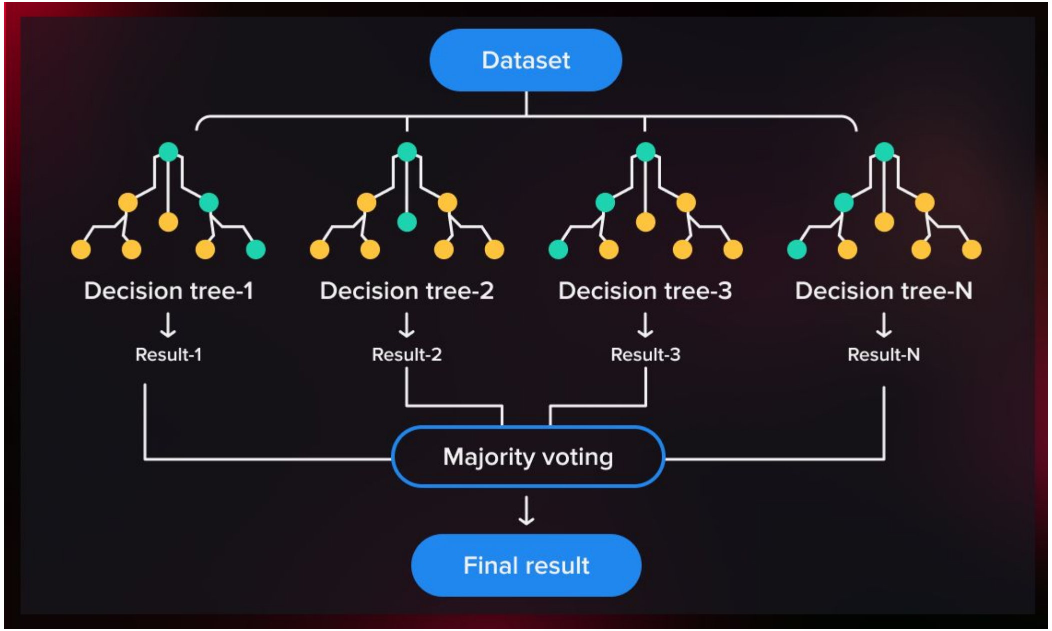
*This is the the definition of entropy for all non-empty classes (p ≠ 0). The entropy is 0 if all samples at a node belong to the same class.

Gini Gain

$$Gini_{gain} = Gini_{parent} - \left(\frac{N_{left}}{N_{total}} Gini_{left} + \frac{N_{right}}{N_{total}} Gini_{right} \right)$$

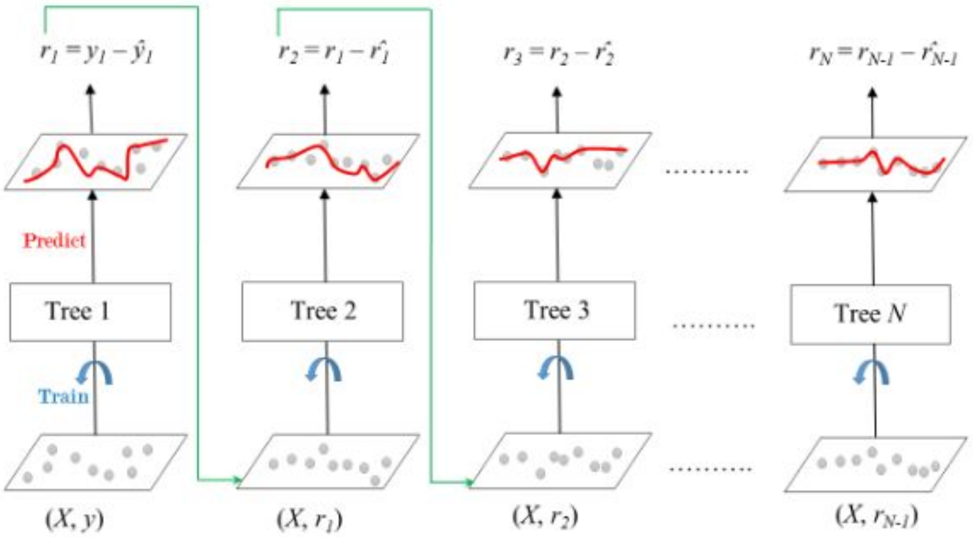
Random Forest

- RandomForestClassifier (sklearn)



Gradient Boosting of Decision Trees

- HistGradientBoostingClassifier (sklearn)
- XGBClassifier
- CatBoostClassifier
- LGBMClassifier



TabPFN Binary Classifier

https://huggingface.co/Prior-Labs/tabpfn_3